Core-to-Margin Radiomic Profiling for Predicting Kidney Cancer Grade from Variable Phase CT

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Study Overview

01 **Clinical Question & cohort**
Benign, low-grade, and high-grade renal masses

02 **Segmentation and Feature Extraction**
Whole mass, kidney/aorta references, core/margin, and 27 patches

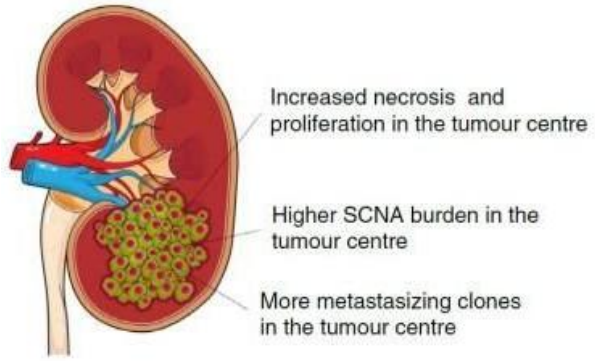
03 **Model Evaluation and Results**
Multiclass and binary classification using patient-grouped five-fold cross validation

04 **Limitations and Conclusions**
Class imbalance, surgical selection, and future multicenter validation

Spatial Heterogeneity in Renal Mass



Heterogeneous attenuation and enhancement within a renal mass [2]



Differences in blood supply and oxygen availability may contribute to biological heterogeneity between the tumor center and margin [3]

Heterogeneity within the tumor

Attenuation and enhancement varies across locations within the same renal mass

Differences from core to margin

Prior research has demonstrated differences in enhancement between central and peripheral tumor regions [1]

Potential clinical relevance

Imaged tumor heterogeneity may contain information related to tumor biology and pathological aggressiveness

Study hypothesis: Spatial distribution of CT heterogeneity, especially core-to-margin differences, may help distinguish benign, low-grade, and high-grade masses

MIDI Lab

[1] Orton et al, Cancer Imaging. 2023:23:76.doi:10.1186/s40644-023-00594-3
[2] CTisus, Johns Hopkins. "Necrotic Renal Cell Carcinoma Extends into the Left Adrenal Bed." Case 6,765.
[3] Zhao Y, et al. *Nature Ecology & Evolution*. 2021;5:1033–1045. doi:10.1038/s41559-021-01456-6. Image credit: The Francis Crick Institute, via Medical Xpress.

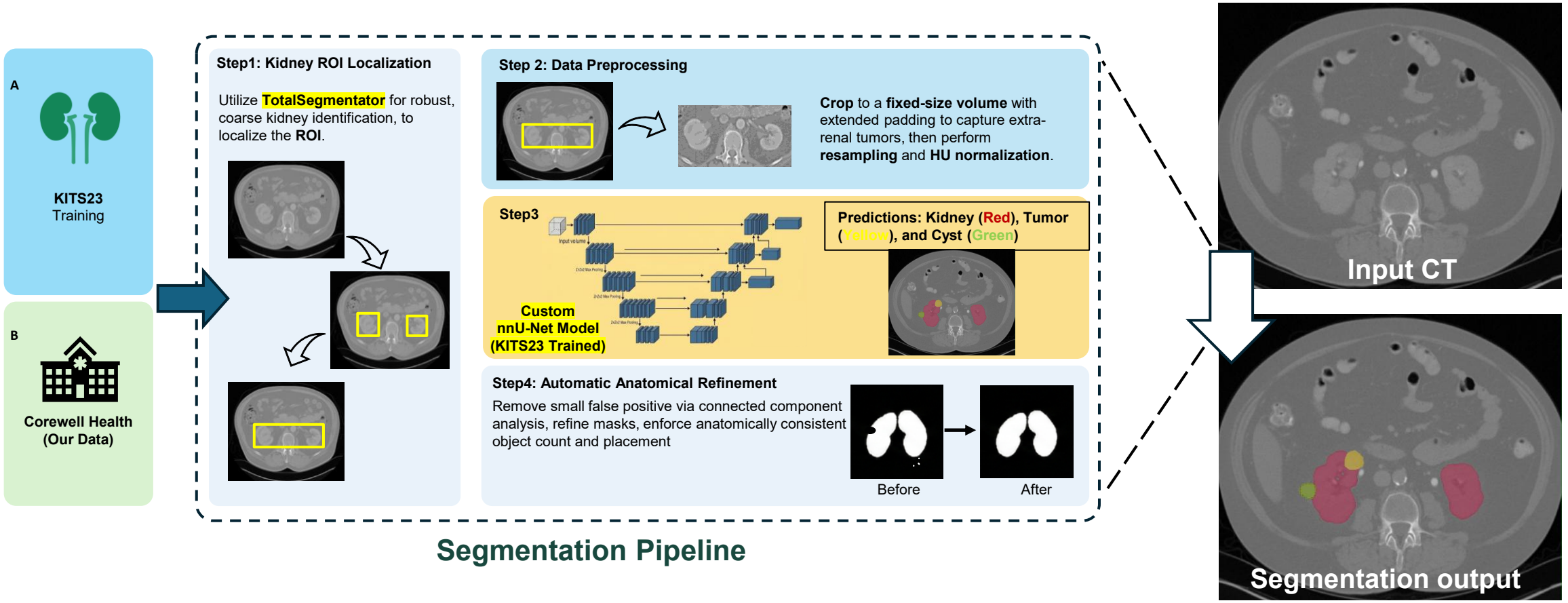
Study Cohort

312 exams with masses characterized by pathology after partial or radical nephrectomy



IRB-approved retrospective study from Corewell Health. Mass volumes: 1–300 cm³.
Low grade: Fuhrman Grade 1 and 2
High grade: Fuhrman Grade 3 and 4

Segmentation and analysis support



Segmentation Pipeline

Whole-body and abdominal CT, with multiple contrast phases.

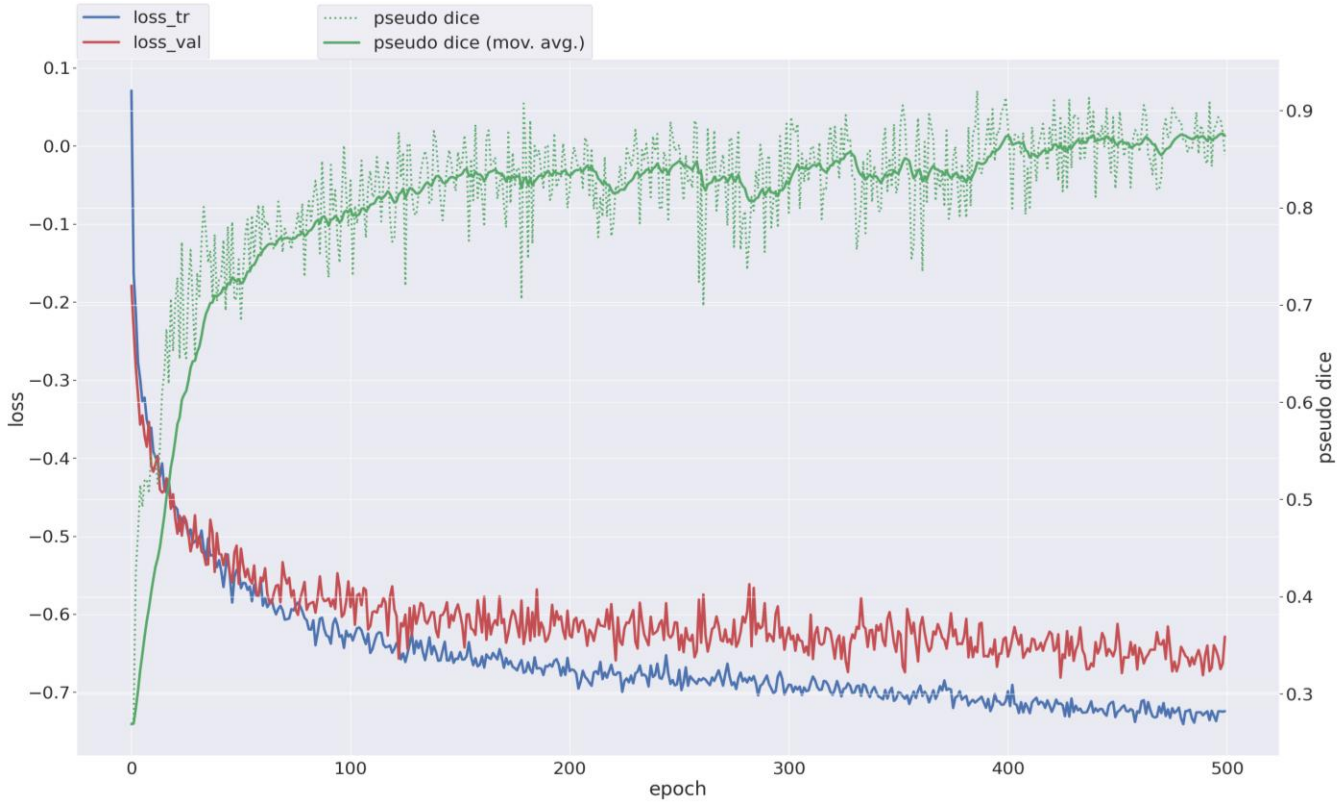
Kidney: red Tumor: yellow Cyst: green

Wasserthal et al. doi:10.1148/ryai.230024 | Isensee et al. doi:10.1038/s41592-020-01008-z

Segmentation training and testing

Training convergence

Loss and pseudo-Dice over 500 epochs

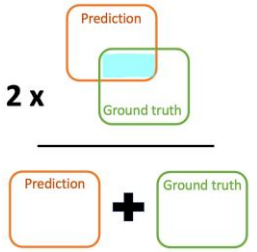


KiTS23 validation

Our segmentation pipeline

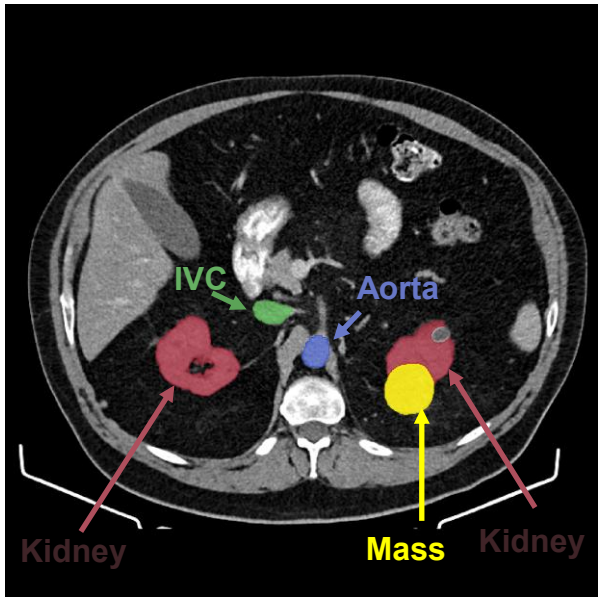
Region	Dice
Kidney	0.9673
Masses	0.8359
Tumor	0.8247
Average	**0.8760**

$$\text{Dice} = \frac{2\,X\,Area\,of\,overlap}{Total\,area} = \frac{2 \times}{}$$

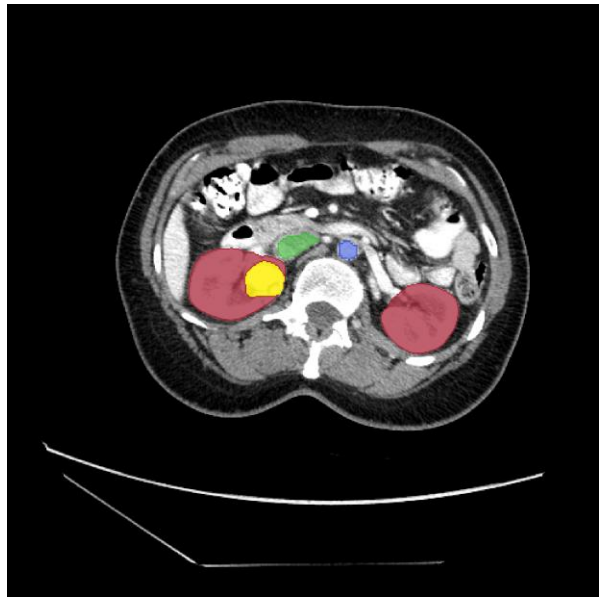


Examples of Segmentations

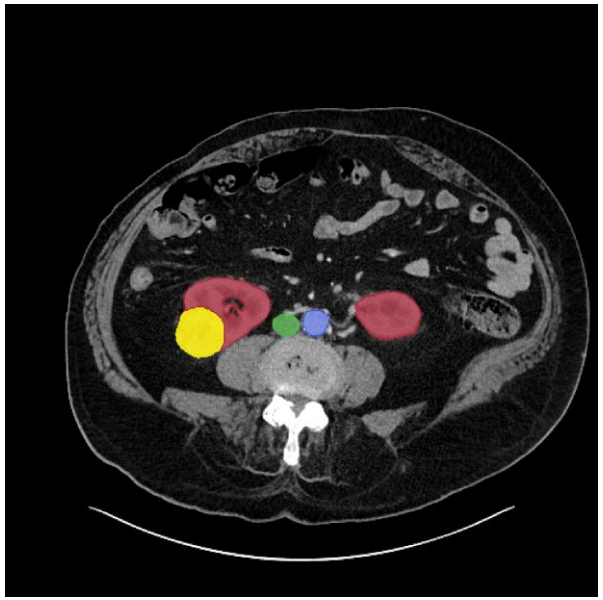
A.



B.



C.

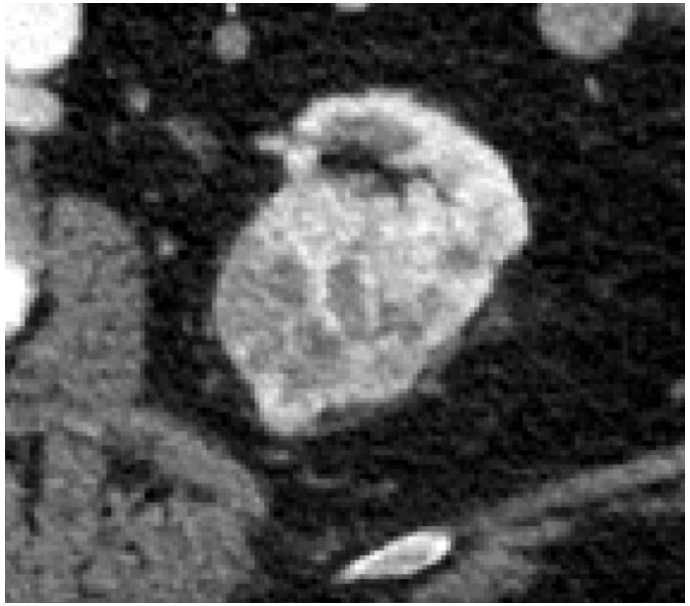


Representative axial slices at the tumor cross-section for **benign (A), low-grade (B),** and **high-grade (C)** renal masses. Segmentation overlays highlight the regions used for radiomics extraction, including the mass (yellow), normal kidney (red), aorta (blue), and inferior vena cava (green).

Experimental comparison

Does retaining spatial information improve classification of high-suspicion renal masses?

High-suspicion renal mass



Feature Extraction Methods:

A 

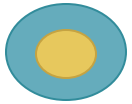
Whole-mass baseline

One feature set for the complete mass

B 

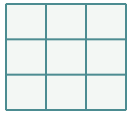
Reference normalization

Mass features with normal kidney & aorta regions to serve as references

C 

Core and margin

Separate central and peripheral feature sets

D 

27 local patches

Local features extracted from a 3 × 3 × 3 patch grid

Primary task: benign, low-grade and high-grade classification within a high-suspicion cohort.

Whole-mass baseline and reference ROIs

Whole-mass baseline



A: Whole-mass Baseline

Normal kidney reference



Aortic reference



B: Kidney and Aorta References

- Reference regions may help with tumor grading

Whole-mass feature selection and model evaluation

Feature selection

107 whole-mass radiomic features extracted from unfiltered CT by **Pyradiomics** [1].

Unsupervised Spearman filtering across all 312 examinations, without classification labels.

Spearman correlation filtering **($|\rho| \geq 0.80$)** reduced feature redundancy

24 features retained after filtering and used across all five folds.

Whole-mass fractal dimension and 3D lacunarity were added, yielding **26 features** across all five folds.

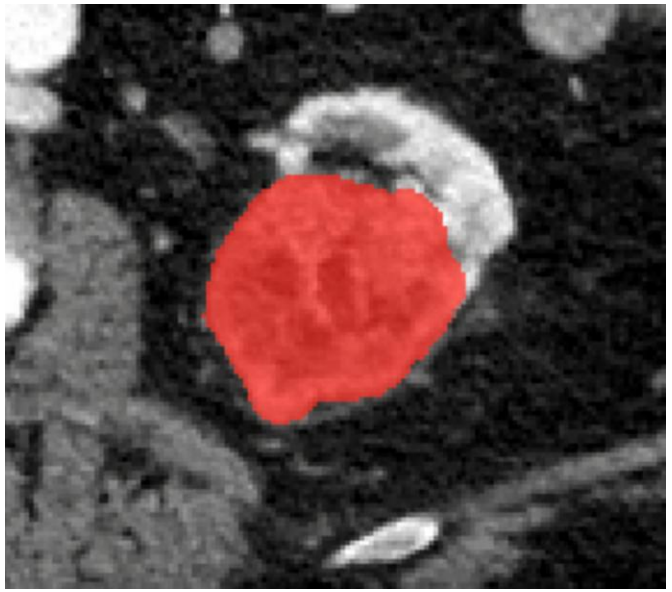
Reference variables	**Normal kidney (3)**	Median, variance, volume
	Aorta (2)	Median, variance

[1] van Griethuysen et al., “Computational Radiomics System to Decode the Radiographic Phenotype,” Cancer Research, 2017.

Fractal features from whole renal mass

We also extracted FD and 3D lacunarity features from whole renal mass.

Whole Renal Mass



Renal mass: red

Fractal dimension (FD)

Characterizes the structural complexity of the renal mass across spatial scales.

3D lacunarity

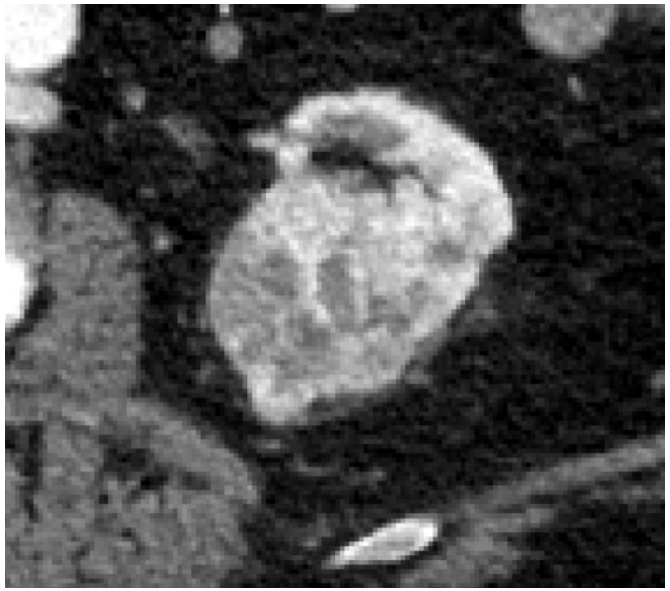
Relative variation of the sampled signal across local 3D windows.

[1]. Concept ImageJ FracL ac, Box Counting documentation. imagej .neti/plugins/fraclac/FLHelp/BoxCounting.htm
[2]. Allain & Cloitre, Phys Rev A. 1991;44:3552. ImageJ FracLac: Lacunarity.

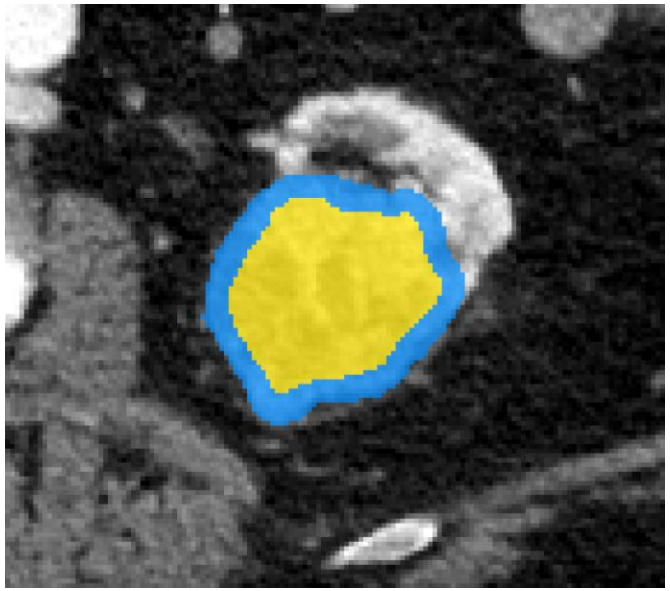
C. Core-to-margin regional profiling

C tests whether central and peripheral descriptors retain information lost by whole-mass pooling.

Original CT



Regional overlay



Core: yellow. Margin: blue.

Features extracted from Core/Margin

Conventional radiomics
Shape, intensity, and texture features

Core and margin features are retained separately.

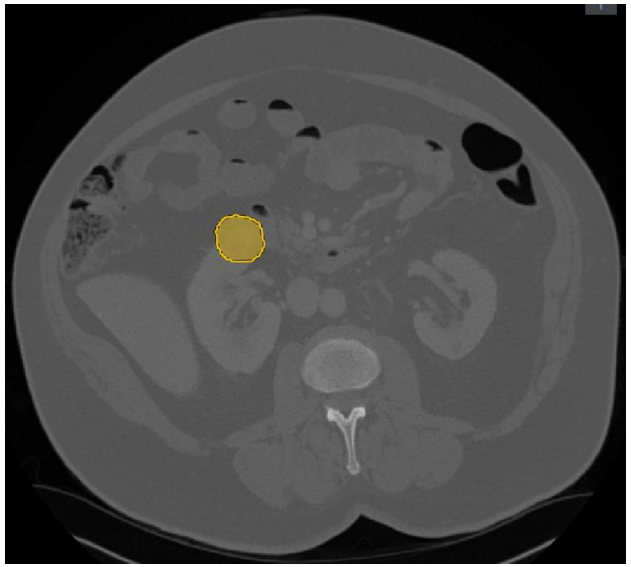
The segmented mass is divided into an inner core and a peripheral margin, each containing approximately 50% of the total mass volume.

Approach C preserves central and peripheral feature sets instead of one pooled summary.

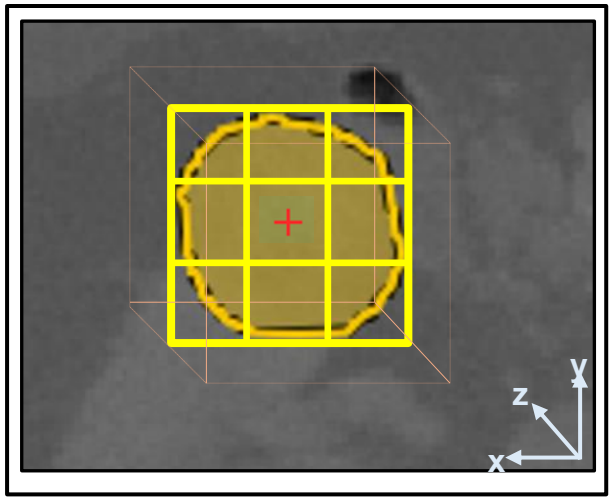
D. Local sampling with 27 patches

Explores finer spatial variation that a two-region representation may miss.

CT context



Local tumor region



Construction

1. Locate the mass centroid
2. Sample a 3 × 3 × 3 local grid
3. Extract mass features per patch

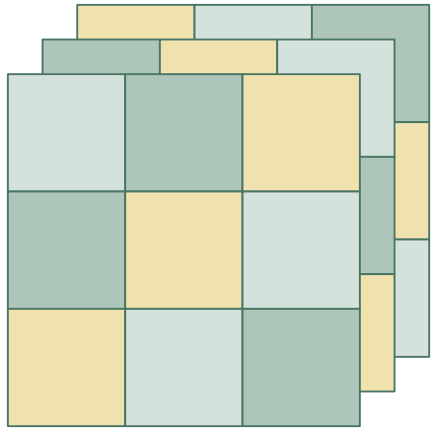
Illustrative 2D layer: 3 × 3. Full grid: 27 patches.

Finer spatial sampling is an alternative to test, rather than an assumed improvement.

D. 27-patch feature aggregation

Initially, four Conventional Radiomics features are extracted from each patch
Aggregation reduces these 4x27 features to 16 features

27 local patches



3 x 3 x 3 grid

4 features per patch

Tumor fill rate
Tumor fraction of the entire patch

Lacunarity
Spatial heterogeneity within each 3D patch

Entropy
Gray-level variability within the patch

Mean CT attenuation
Mean HU of tumor voxels in the patch

4 summaries per feature

Use the 27 values of one feature to calculate its four summaries.
Repeat for each feature.

→ **4 fill-rate summaries**
Max / top-5 mean / top-5 SD / high-low

→ **4 lacunarity summaries**
Max / top-5 mean / top-5 SD / high-low

→ **4 entropy summaries**
Max / top-5 mean / top-5 SD / high-low

→ **4 mean-HU summaries**
Max / top-5 mean / top-5 SD / high-low

High-low: mean of highest 5 minus mean of lowest 5, for the same feature.

4 fill-rate + 4 lacunarity + 4 entropy + 4 mean-HU summaries = 16 inputs

Model training and validation

Selected feature set

Separate fit for each of A–D

→

Three-class model

CatBoost

Same classifier [1] for all feature sets

→

Predicted class

Benign
Low-grade
High-grade

Patient-grouped five-fold cross-validation

312 examinations; folds grouped by patient.

4 folds for training and 1 for validation.

Same five splits reused for all feature sets.

Five fitted models and five validation AUCs per feature set.

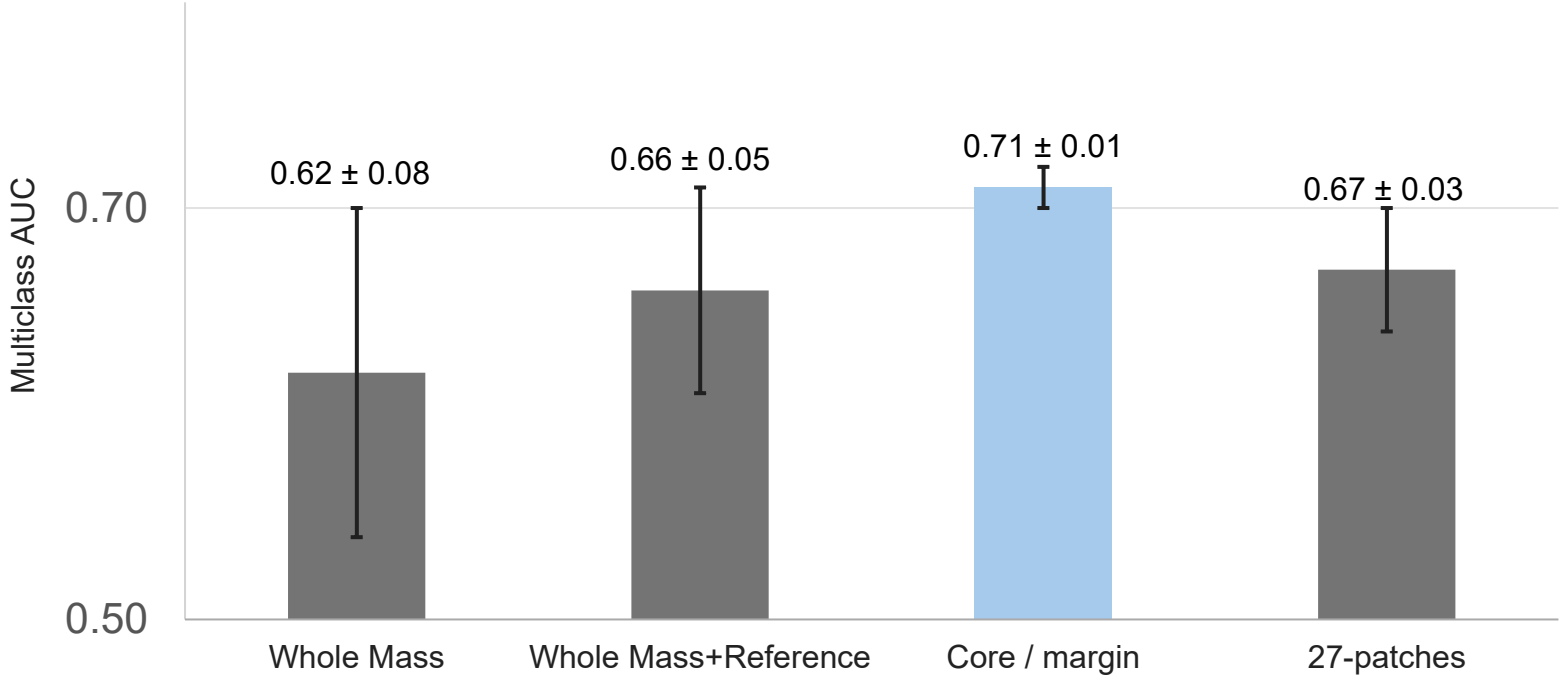


Mean AUC (SD) across the five validation folds

[1] Prokhorenkova et al., “CatBoost: Unbiased Boosting with Categorical Features,” NeurIPS, 2018..

Multiclass (Benign, Low-, High- Grade) discrimination across approaches

Performance results compare whole-region, +reference and spatial feature approaches.



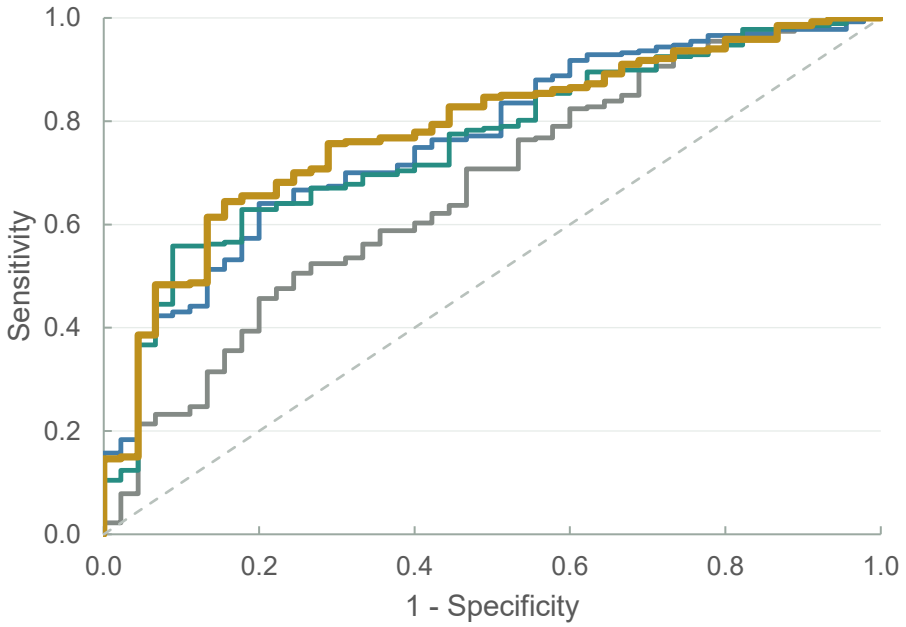
The four result groups correspond to approaches A–D defined on slide 8.

Core/margin analysis achieved the highest AUC. The 27-patches approach improved on whole-mass radiomics, achieving discrimination comparable to adding reference features.

Binary discrimination analyses: Benign vs malignant

45 benign / 267 malignant. Same random seed and patient-grouped five-fold CV across all models. Balanced CatBoost.

Pooled out-of-fold ROC



Five-fold mean ± SD

Feature set	AUC	Accuracy
Whole mass	0.69 ± 0.08	0.80 ± 0.05
Whole mass + kidney/aorta	0.76 ± 0.08	0.83 ± 0.03
Core/margin	0.77 ± 0.06	0.82 ± 0.02
27-patches	**0.79 ± 0.06**	0.83 ± 0.04

Binary classification showed a similar pattern, with higher AUCs after adding kidney/aorta and regional features, and the highest mean AUC with 27-patch features.

Study limitations and future directions

Study purposely enrolled only high-suspicion lesions that resulted in nephrectomy
Class imbalance and surgical selection should be considered when interpreting these findings

Class distribution in the surgical cohort

312 renal masses

Low-grade tumors account for 61% of cases



45	190	77
Benign	Low-grade	High-grade

Limitations and next steps

Class imbalance

Benign and high-grade masses are less represented.

Evaluate per-class recall and balanced accuracy, and expand recruitment of these groups.

Surgical selection

This cohort includes masses selected for nephrectomy.

Validate across centers, including biopsy and surveillance cohorts.

Validation priorities Multicenter validation | Acquisition and volume robustness | Calibration and clinical utility

Conclusions

The study tests whether preserving spatial information improves renal mass classification.

01 Clinical task

Classifying high-suspicion renal masses as benign, low-grade, or high-grade, with additional benign-versus-malignant analysis.

02 Observed finding

In the original multiclass analysis, kidney/aorta reference increased AUC from 0.62 to 0.66. Core/margin achieved the highest reported AUC of 0.71, while 27-patches yielded 0.67.

In the updated binary analysis, reference features increased mean AUC from 0.69 to 0.76, followed by 0.77 with core/margin and 0.79 with 27-patches features.

03 Next steps

Assess class-specific performance, calibration, and generalizability through multicenter validation.

Preserving spatial information is promising; clinical benefit remains to be demonstrated

Selected methodological references

Primary sources for segmentation, feature interpretation and reproducible reporting.

1 Wasserthal et al. TotalSegmentator. Radiology: AI. 2023.
doi:10.1148/ryai.230024

2 Isensee et al. nnU-Net. Nature Methods. 2021.
doi:10.1038/s41592-020-01008-z

3 Allain & Cloitre. Characterizing lacunarity. Physical Review A. 1991.
doi:10.1103/PhysRevA.44.3552

4 Zwanenburg et al. Image Biomarker Standardization Initiative. Radiology. 2020.
doi:10.1148/radiol.2020191145

5 ImageJ FracLac. Box counting and sliding-box documentation.
imagej.net/ij/plugins/fraclac/FLHelp/BoxCounting.htm

6 KiTS23. Kidney, tumor and cyst segmentation resource.
kits-challenge.org/kits23/



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START THE CURE STARTS HERE

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